

KB002 - Authentication and Login Failure Troubleshooting SOP

Document Metadata

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1. Overview

The Authentication Service manages user identity verification, session creation, access token generation, and authorization workflows across customer-facing applications.

The service is responsible for:

- User login
- User registration
- Session validation
- JWT token generation
- OAuth authorization
- Multi-factor authentication
- Password reset workflows

Authentication incidents usually cause direct customer impact because users cannot access business services.

Common business impacts:

- Users unable to login

- Increased failed login attempts
 - Application access failures
 - API authorization errors
 - Customer support ticket increase
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2. Common Incident Symptoms

User Symptoms

Customers may report:

- Unable to login
 - Login page continuously loading
 - Session expires immediately after login
 - Password reset emails not received
 - Multi-factor authentication failures
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Application Symptoms

Common HTTP errors:

```
HTTP 401 Unauthorized
HTTP 403 Forbidden
HTTP 500 Internal Server Error
HTTP 503 Authentication Service Unavailable
```

Example application logs:

```
ERROR AuthService
Authentication failed.
Exception:
TokenValidationException
Message:
JWT signature verification failed
```

Another common error:

```
OAuthProviderException:
Unable to retrieve authorization response from identity provider
```

3. Monitoring Indicators

Before starting RCA, verify authentication metrics.

Authentication Metrics

Metric	Warning Threshold	Critical Threshold
login_failure_rate	>10%	>30%
authentication_latency_p95	>2 sec	>5 sec
token_validation_errors	>100/min	>1000/min
active_sessions_drop	>20%	>50%

Infrastructure metrics:

Monitor:

- Authentication API latency
- Database availability
- Cache availability
- Identity provider response time
- Certificate expiration
- Token validation errors

4. Common Root Causes and Resolution Steps

RCA Pattern 1: JWT Signing Certificate Expired

Description

Authentication services use certificates/private keys to sign and validate JWT tokens.

If the certificate expires or becomes invalid, generated tokens cannot be trusted by downstream services.

Symptoms

User impact:

- All users suddenly logged out
- Login succeeds but APIs fail
- Sessions expire immediately

Application logs:

```
JWTValidationException:  
Token signature verification failed  
  
CertificateExpiredException:  
Signing certificate expired
```

Validation Steps

Step 1:

Check certificate expiration:

```
openssl x509 \  
-in auth_certificate.pem \  
-noout \  
-dates
```

Expected:

```
Certificate expiry date should be greater than current date
```

Step 2:

Verify token validation service.

Check:

```
jwt.validation.success.rate  
jwt.validation.failure.count
```

Immediate Resolution

1. Rotate JWT signing certificate.
 2. Deploy updated certificate.
 3. Restart Authentication Service instances.
 4. Force refresh authentication metadata cache.
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Permanent Fix

Implement:

- Automated certificate rotation
 - Expiry monitoring alerts
 - Certificate renewal pipeline
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RCA Pattern 2: Authentication Database Connectivity Failure

Description

Authentication Service depends on the user database for:

- User lookup
- Password verification
- Permission retrieval
- Account status validation

Database issues can cause login failures.

Symptoms

Customer impact:

Login requests timing out

Users unable to authenticate

Application logs:

DatabaseConnectionException

Unable to acquire connection from pool

or

UserRepositoryTimeoutException

User lookup exceeded timeout

Validation Steps

Check database connectivity:

```
SELECT status
```

```
FROM database_health;
```

Check active connections:

```
SELECT COUNT(*)  
FROM active_sessions;
```

Review:

```
database.connection.pool.usage  
database.response.time
```

Immediate Resolution

Actions:

1. Restart unhealthy database connections.
 2. Increase connection pool temporarily.
 3. Scale database read replicas.
 4. Restart authentication workers.
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Permanent Fix

- Optimize user lookup queries
 - Add database indexes
 - Implement caching layer
 - Configure connection monitoring
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RCA Pattern 3: Redis / Session Cache Failure

Description

Authentication systems commonly store:

- Active sessions
- Refresh tokens
- User permissions

inside distributed cache systems.

If cache becomes unavailable, authentication failures may occur.

Symptoms

Examples:

```
Users randomly logged out
```

```
Sessions disappearing
```

```
Multiple login prompts
```

Logs:

```
RedisConnectionException
```

```
Unable to connect to cache cluster
```

or

```
SessionNotFoundException
```

Validation

Check cache availability:

```
redis-cli ping
```

Expected:

```
PONG
```

Monitor:

```
cache_hit_ratio
```

```
cache_connection_errors
```

```
cache_memory_usage
```

Resolution

Immediate:

- Restart unhealthy cache nodes
- Failover to replica node
- Clear corrupted session entries

Permanent:

- Configure Redis clustering
- Enable automatic failover
- Add cache health monitoring

RCA Pattern 4: External Identity Provider Failure

Description

Authentication service may depend on external providers:

Examples:

- Enterprise SSO provider
- OAuth provider
- Identity federation service

Provider outages can block authentication.

Symptoms

Only specific users impacted:

Example:

Corporate users cannot login

Normal email/password users working

Logs:

OAuthTimeoutException

Identity provider response timeout

Validation Steps

Check:

identity_provider_latency

oauth_success_rate

oauth_callback_errors

Verify provider health dashboard.

Resolution

Short term:

- Enable fallback login mechanism

- Increase provider timeout temporarily
- Retry failed authorization requests

Long term:

- Add redundant identity provider
 - Implement circuit breaker
 - Improve provider monitoring
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RCA Pattern 5: Recent Deployment Configuration Issue

Description

Authentication failures frequently occur after:

- Application deployment
- Configuration changes
- Security policy updates

Common mistakes:

- Wrong OAuth redirect URL
 - Incorrect secrets
 - Missing environment variables
 - Invalid token configuration
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Detection

Compare:

```
incident_start_time
```

```
vs
```

```
latest_deployment_time
```

Check deployment history:

```
auth-service:v5.2.1 deployed
```

Validation

Review configuration:

```
auth:
```

```
tokenExpiry: 3600  
issuer: auth.company.com  
oauthRedirectURL:  
https://application.company.com/callback
```

Check error rate before/after release.

Resolution

1. Stop rollout.
 2. Restore previous configuration.
 3. Rollback authentication deployment.
 4. Validate login success rate.
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5. Incident Investigation Checklist

Follow this sequence during authentication outage.

Step 1: Identify Impact

Questions:

- Are all users affected?
 - Specific region?
 - Specific login method?
 - Specific application?
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Step 2: Check Recent Changes

Verify:

- Deployments
 - Certificate updates
 - Security policy changes
 - Infrastructure changes
-

Step 3: Validate Dependencies

Check:

- User database
 - Redis cache
 - Identity provider
 - Network connectivity
-

Step 4: Apply Recovery

Possible actions:

- Rotate certificate
 - Restart service
 - Restore configuration
 - Rollback deployment
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6. Escalation Matrix

Issue Type	Escalation Team
Token / JWT failures	Identity Platform Team
Database issues	Database Engineering
Cache failures	Infrastructure Team
SSO provider outage	Security Engineering
Deployment defect	Application Development Team

7. Related Historical Incidents

INC-20341

Issue:

Users unable to login after certificate expiry.

Root Cause:

JWT signing certificate expired.

Resolution:

Rotated signing certificate and restarted authentication services.

INC-20781

Issue:

Random session logout reported by customers.

Root Cause:

Redis cache node failure.

Resolution:

Performed cache cluster failover.

INC-21109

Issue:

Authentication failures after deployment.

Root Cause:

Incorrect OAuth redirect configuration.

Resolution:

Rolled back configuration change.

8. RCA Generation Template

Authentication RCA reports should contain:

Root Cause

Technical cause responsible for authentication failure.

Evidence

Include:

- Logs
- Metrics
- Related incidents
- KB references

Resolution Applied

Immediate recovery actions.

Preventive Actions

Long-term reliability improvements.

Example:

Root Cause:

Authentication outage occurred because JWT signing certificate expired.

Evidence:

- Token validation errors increased suddenly.
- Logs showed CertificateExpiredException.
- Similar incident INC-20341 identified.

Resolution:

Rotated certificate and restarted Authentication Service.

Preventive Action:

Implemented automated certificate expiry monitoring.

End of Document